

## @cta-optimizer

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*"...mistral:7b sometimes returned 'Variant 1' with a capital V and a space instead of 'variant\_1', and the template threw a KeyError. The fix normalises all LLM output keys to snake\_case before any agent returns."*

### @cta-optimizer

[agents/cta\\_optimizer.py:63-76](#)

```
63         # Normalize keys – models sometimes use "Variant 1", "variant1", etc.
64         key_map = {}
65         for k in parsed:
66             normalized = k.lower().replace(" ", "_")
67             if "1" in normalized:
68                 key_map["variant_1"] = parsed[k]
69             elif "2" in normalized:
70                 key_map["variant_2"] = parsed[k]
71
72         if "variant_1" in parsed and "variant_2" in parsed:
73             return parsed
74         if key_map:
75             return key_map
76         raise ValueError(f"Unexpected CTA response keys: {list(parsed.keys())}")
```

**What it shows:** The fix: lower-case + de-space the model's variant keys and remap to variant\_1 / variant\_2, so a "Variant 1" response no longer throws a KeyError in the template.

Source: [agents/cta\\_optimizer.py:63-76](#) · image rendered by [build\\_snippets.py](#)